

Epistemic Memory: A Local, LLM-Free Memory Layer with Operational Self-Knowledge for AI Agents

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Abstract

Persistent memory is what separates a stateless chatbot from an assistant that knows you, and a wave of systems (MemGPT/Letta, Mem0, Zep, A-Mem) now gives agents long-term memory. Two assumptions run through much of this literature: that a cloud LLM performs the cognitive work (extraction, summarization, conflict resolution, importance scoring), and that per-memory reliability (confidence, provenance) is inert metadata that is written but never read back. We take the opposite stance on both. We present **Memory Layer**, an open-source system whose entire maintenance loop of decay, spaced-repetition reinforcement, episodic→semantic consolidation, contradiction detection, and a six-check self-audit runs fully on-device with no LLM call and no network, over a local embedding model and an on-disk SQLite + FAISS store. On top of this we make *belief operational*: every memory carries a graded confidence and a four-state *epistemic status* (VERIFIED, INFERRED, UNCERTAIN, CONTRADICTED) that is assigned at formation, revised by contradiction, correction, and supersession, and folded into the retrieval score, so an unreliable memory is demoted even when it matches the query well. The contribution is not a new retrieval algorithm (each mechanism has well-known antecedents) but their *integration under a strict local/LLM-free constraint* and the *operationalization of self-knowledge*, which we characterize on a laptop: forgetting dynamics, the ranking effect of the epistemic modifier, end-to-end contradiction handling, self-audit precision on planted issues, and recall latency/footprint. We do *not* claim state-of-the-art long-horizon QA accuracy. We further show, via a fair ablation that toggles *only* the epistemic modifier, that folding belief into ranking eliminates confident answers drawn from low-trust memories (33.3%→0%) without sacrificing answerable coverage; we frame belief propagation through consolidation and external (LongMemEval) benchmarking as the next steps, and release reproduction scripts for every figure.

1 Introduction

An LLM is, in isolation, a pure function of its prompt: it does not remember the user it spoke to yesterday, the correction it received an hour ago, or the document it read last week. For one-shot answering this is fine; for *agents* that act over long horizons it is disqualifying. An agent that cannot remember cannot personalize, accumulate expertise, honor a correction, or keep a consistent account of the world. Persistent memory is thus a precondition for agency, and the field has responded with the *memory layer*: an external store beside the model that captures salient information and retrieves it on demand [1, 2, 3, 4, 5, 6].

Two assumptions run through much of this work (§2 surveys recent exceptions). First, **a cloud LLM does the cognitive work**: extraction, summarization, conflict resolution, and importance judgments are delegated to an API model, coupling memory quality to network access, per-token cost, and shipping a user’s most personal data off-device. Second, **reliability**

is decoration: systems may *store* a confidence or a source, but that signal rarely changes what is retrieved: it is written and never read back into ranking. This paper takes the opposite stance on both. We ask: how much self-maintaining memory behavior survives if we forbid the LLM and the network entirely, and what happens if a memory’s self-assessed reliability is made to actually govern retrieval?

Shared memory raises the stakes. A single user runs many agents (a coding copilot, an email assistant, a research agent), and teams run fleets of agents over a common store. Portable, shared memory lets a correction taught to one agent benefit all of them. But a memory written by agent *A* and read by agent *B* must carry its reliability with it, or *B* has no way to tell a user-verified fact from a stale guess. Shared memory without an epistemic signal is a channel for confidently propagating error.

Contributions. We are explicit that no single mechanism here is new: decay is Ebbinghaus/MemoryBank, consolidation and reflection are Generative Agents, confidence/justification tracking is classical truth maintenance, and the self-audit mirrors the “LLM Wiki” lint idea. The contribution is the integration under a hard constraint and the operationalization of self-knowledge, demonstrated and measured on-device:

1. **C1 — A fully local, LLM-free maintenance loop.** Decay, reinforcement, consolidation (with an extractive fallback), contradiction detection/supersession, and a six-check self-audit run with zero LLM calls and zero network I/O (§4).
2. **C2 — Operational self-knowledge.** A stored confidence and a four-state epistemic status are assigned at formation and revised across the memory lifecycle, and are multiplied into the composite retrieval score via an *epistemic modifier*; a memory that is unreliable is demoted even when it matches the query. A `lint` pass audits the store’s own contradictions, staleness, orphans, and duplicates (§3–4).
3. **C3 — On-device characterization and a fair ablation.** We measure forgetting dynamics, the ranking effect of the modifier, contradiction handling, audit precision, and latency/footprint on a laptop; toggling *only* the epistemic signal, we show it reduces confident answers drawn from low-trust memories at matched answerable coverage (§5.6). All figure- and number-generating scripts are released (§5).

We do not claim improved accuracy on LoCoMo/LongMemEval; that requires head-to-head runs we have not performed. We frame belief propagation through consolidation and external benchmarking as next steps (§6), and flag all limitations plainly (§7).

2 Background and Related Work

Biologically-inspired memory. MemoryBank [5] applies the Ebbinghaus forgetting curve to LLM memory; Generative Agents [6] introduce a memory stream with recency/importance/relevance retrieval and periodic *reflection* that abstracts observations. Our decay, spaced-repetition reinforcement, and episodic→semantic consolidation follow this lineage; our twists are type- and abstraction-level stability multipliers and an LLM-free extractive consolidation fallback.

Agent memory systems. Mem0 [2] delegates add/update/delete memory operations to an LLM and reports large token/latency savings over full context; Zep/Graphiti [3] maintain a bi-temporal knowledge graph; MemGPT/Letta [1] give the model tool-callable, self-editing memory; A-Mem [4] evolves Zettelkasten-style linked notes. These are powerful but LLM- and often cloud-dependent. We occupy a different point in the design space: local, deterministic

maintenance with the LLM optional. Retrieval-augmented generation [7] is the substrate these specialize.

Knowledge organization and self-audit. The “LLM Wiki” pattern proposes auto-maintained per-entity pages plus a *lint* operation for knowledge health, and GraphRAG [8] builds community summaries over a graph. Our self-audit is a direct, LLM-free re-implementation of the lint idea.

Truth maintenance, provenance, and uncertainty. Tracking justifications and belief status, and revising them as premises change, is classical: truth-maintenance systems [9], assumption-based TMS [10], and AGM belief revision [11]. Data provenance [12] characterizes the origin and derivation of data. Calibration [13] and selective prediction [14] study when a model should decline. Our epistemic status and confidence are a lightweight, embedding-era instance of these ideas; the novel part is wiring them into ranking and surfacing them through a self-audit. Long-horizon benchmarks LongMemEval [15] and LoCoMo [22] include explicit *knowledge-update* and *abstention* abilities—evidence the field treats belief revision and calibrated refusal as first-class.

Concurrent systems (2025–2026). The design space this paper occupies is forming rapidly, and several concurrent works overlap individual components. MemX [16] is a local-first memory system with a low-confidence rejection rule that suppresses spurious recalls (local-first plus score-based abstention, though its embeddings are served via an OpenAI-compatible API and it does not model discrete epistemic states). Hindsight [17] maintains separate epistemic networks whose beliefs carry evolving confidence scores. Governed Memory [18] addresses memory governance and quality feedback loops for multi-agent enterprise workflows. TOKI [19] gives a bitemporal operator algebra for contradiction resolution; Eywa [20] grounds long-term memory in provenance; and MemAudit [21] audits agent memory post-hoc for poisoning. Each of these validates one pillar of our design: confidence-gated recall, graded belief, governance, contradiction handling, provenance, and auditing respectively.

Positioning. Against this backdrop our claim is deliberately narrow. We contribute, in one open system: (a) a maintenance loop (decay, consolidation, contradiction handling, self-audit) that is strictly local *and* LLM-free, including the embeddings; (b) a *four-state epistemic status revised across the full memory lifecycle* (formation, contradiction, correction, supersession) rather than a single confidence scalar; (c) that status and confidence folded multiplicatively into the retrieval score; and (d) a *controlled ablation* (§5.6) isolating that the epistemic signal itself, not merely the presence of an abstention rule, reduces confident errors. To our knowledge no concurrent system publishes that combination, and in particular none reports the component-isolating ablation in (d).

3 The Epistemic Memory Model

3.1 The epistemic memory atom

Definition 1 (Epistemic memory). An epistemic memory is a tuple

$$m = \langle c, \mathbf{e}, \kappa, \sigma, s, \iota, \text{cur}, P \rangle,$$

where c is the content, $\mathbf{e}$ its embedding, $\kappa \in [0, 1]$ a *confidence*, σ a discrete *epistemic status* drawn from {VERIFIED, INFERRED, UNCERTAIN, CONTRADICTED}, $s \in [0, 1]$ a decaying *strength*, $\iota \in [0, 1]$ an *importance*, cur a *currency* flag (false once superseded), and P a pointer into the provenance log.

The pair (κ, σ) is the load-bearing addition: κ is a continuous trust used for ranking and thresholding; σ is a coarse, human-legible label used for ranking penalties and audit.

3.2 Belief assignment at formation

A new memory’s belief is set from the *nature of its source*, not its content:

$$(\kappa_0, \sigma_0) = \begin{cases} (0.9, \text{VERIFIED}) & \text{user correction,} \\ (0.8, \text{INFERRED}) & \text{document/URL source,} \\ (0.7, \text{INFERRED}) & \text{extracted semantic fact,} \\ (0.5, \text{INFERRED}) & \text{otherwise.} \end{cases} \quad (1)$$

These are heuristic priors, not calibrated probabilities (§7).

3.3 Belief revision over the lifecycle

Principle 1 (Supersession). If a new memory is an *update* of a sufficiently similar prior one (cosine plus lexical-overlap test), the old memory is snapshoted, marked non-current, attenuated in strength, set to $\sigma = \text{CONTRADICTED}$ with its confidence floored, and removed from the primary index while remaining queryable; a **SUPERSEDED** edge is recorded. The store is append-only.

Principle 2 (Soft contradiction). If a new memory conflicts but is not a clean update, a **CONTRADICTS** edge is recorded, both remain current, and both are penalized at retrieval and surfaced by the self-audit until resolved.

Principle 3 (Correction). An explicit user correction creates a **VERIFIED**, high-confidence memory, supersedes the old one, and logs a **CORRECTED** provenance edge.

Principle 4 (Decay). Unrecalled memories lose strength on an Ebbinghaus-style curve; recall reinforces them. Decay attenuates *retrievability* (s), not *belief* (κ): a fact can be firmly believed yet hard to recall.

The framing we emphasize is that belief is a quantity that is *set once and then transported and revised* through every operation a memory undergoes: supersession demotes it, correction promotes it, contradiction marks it, and retrieval reads it, with an auditable record at each step. It is this lifecycle-wide use, not the mere presence of a score, that distinguishes epistemic memory from a vector store with a confidence column.

4 System: Memory Layer

Memory Layer stores every memory in a local SQLite database (source of truth, WAL mode) with a FAISS index for dense retrieval and auxiliary entity-graph and full-text (FTS5) indices. Embeddings default to a local **all-mpnet-base-v2** model (768-d); an optional local cross-encoder (**ms-marco-MiniLM-L-6-v2**) reranks candidates. After a one-time model download at install, nothing in the default path contacts a network service.

Retrieval is a two-stage cascade: FAISS + FTS + entity-graph expansion produce candidates, scored by a composite of semantic similarity (dominant), lexical, phrase, entity, temporal, and type signals, then optionally reranked. On top of this *relevance* score we apply lifecycle *modifiers* (strength, importance, recency/history, supersession) and the *epistemic modifier* below.

Maintenance runs in the background and on demand, entirely LLM-free: decay with stability scaling by memory type (procedural $5\times$, semantic $3\times$, episodic $1\times$), abstraction level, importance, and $\ln(2 + \text{access_count})$ (spaced repetition); consolidation of related episodes into semantic memories (true synthesis with an LLM; extractive keyword summarization without one); and contradiction detection driving the supersession and soft-contradiction principles of §3.

4.1 The epistemic modifier (the read path)

This is where self-knowledge becomes operational. For each candidate the composite score is multiplied by

$$\text{epi}(m) = (W_f + (1 - W_f) \kappa) \cdot \pi(\sigma), \quad W_f = 0.7, \quad (2)$$

with status penalty $\pi(\text{VERIFIED}) = \pi(\text{INFERRED}) = 1.0$, $\pi(\text{UNCERTAIN}) = 0.85$, $\pi(\text{CONTRADICTED}) = 0.5$. Thus confidence maps to a multiplier in $[W_f, 1]$ and the two “bad” statuses apply an additional categorical penalty. A flag (`use_epistemic`) disables the modifier for ablation. This turns confidence from write-only metadata into a retrieval signal (measured in §5.2).

4.2 Self-audit (`lint`)

A single deterministic pass runs six checks over the store: unresolved contradictions (both memories current), stale memories (old and never accessed), orphans (no links/entities), outdated knowledge pages, entity coverage gaps, and near-duplicates (cosine > 0.95). It returns a structured report for the CLI or UI. No LLM is involved, so “memory hygiene” becomes a measurable, repeatable operation.

5 Experiments

All experiments run on a single machine with local embeddings, the local reranker when available, **LLM enrichment disabled**, and no network. §5.1–5.5 are produced by the released characterization script (seed = 42); the fair ablation of §5.6 by the released ablation script (deterministic). Both are in the accompanying repository.

5.1 Forgetting dynamics

Figure 1 plots retained strength versus time since last recall for each memory type and for a reinforced (recalled $5\times$) episodic memory. As designed, episodic memories fade fastest (reaching the strength floor within ~ 8 days), semantic and procedural persist progressively longer, and reinforcement markedly flattens the episodic curve—reproducing the qualitative Ebbinghaus + spaced-repetition behavior with no LLM.

5.2 Self-knowledge is operational

We fix a memory’s query relevance and sweep its stored confidence across each epistemic status, measuring the resulting composite retrieval score with the modifier **on** versus a baseline with it **off** (relevance only, score 0.5945). Figure 2 shows the score rising with confidence and the UNCERTAIN/CONTRADICTED curves sitting strictly below VERIFIED/INFERRED (at confidence 1.0: verified 0.5945, uncertain 0.5053, contradicted 0.2972). Table 1 reports the modifier $\text{epi}(m)$ itself, independent of relevance.

5.3 End-to-end contradiction handling

We store a fact, then a contradicting update, and query for it. The old memory’s status becomes CONTRADICTED (confidence floored to 0.3, marked non-current), the new memory *is* returned

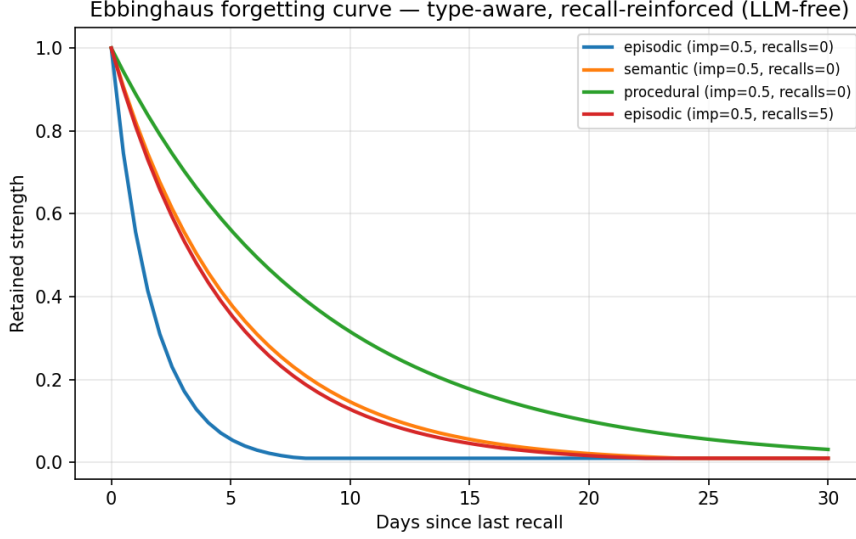


Figure 1: On-device forgetting dynamics by memory type. Episodic appears twice deliberately: once never recalled (fastest fade) and once recalled $5\times$, showing that recall reinforcement flattens the curve. Type stability multipliers: episodic $1\times$, semantic $3\times$, procedural $5\times$; recall raises stability via $\ln(2 + \text{access_count})$.

Table 1: The epistemic modifier $\text{epi}(m)$ of Eq. (2), by status and confidence.

status	$\kappa = 0.0$	$\kappa = 0.5$	$\kappa = 1.0$
VERIFIED	0.70	0.85	1.00
INFERRED	0.70	0.85	1.00
UNCERTAIN	0.595	0.722	0.85
CONTRADICTED	0.35	0.425	0.50

by recall while the stale one is *suppressed* (recall returns new = true, returns old = false), and the supersession link is recorded for audit.

5.4 Self-audit precision

On a fresh store seeded with healthy filler plus *planted* issues, `lint` recovers them (Table 2). Planted issues are controlled—explicit CONTRADICTS links, back-dated never-accessed memories, and an embedding-identical near-duplicate—so this measures detection mechanics, not real-world prevalence.

Table 2: Self-audit detection on planted issues (`lint`).

issue type	planted	caught
contradiction	1	1
stale	3	3
duplicate	1	1

5.5 Latency and footprint

Storing 400 memories and issuing 80 recalls on a laptop (local embeddings + local cross-encoder reranker, no LLM): recall latency p50 = 78.6 ms, p95 = 190.5 ms (mean 95.1 ms); store

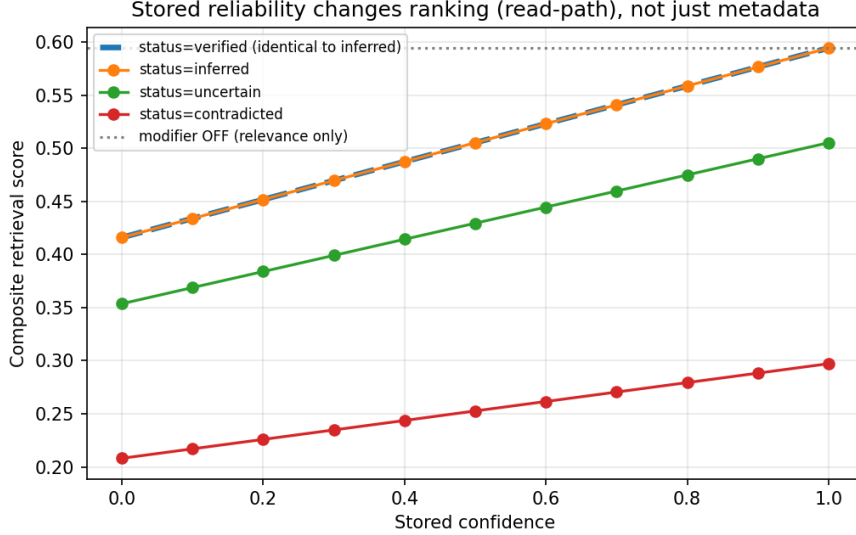


Figure 2: Composite retrieval score at fixed relevance as stored confidence varies, per epistemic status. The VERIFIED curve (dashed) coincides exactly with INFERRED, since both carry status penalty 1.0. Identical relevance, different reliability, different ranking—the concrete sense in which self-knowledge is operational.

throughput 27.8 ms/memory; on-disk footprint ≈ 4.4 MB (SQLite) + ≈ 4.8 MB (FAISS). The system is comfortably interactive on consumer hardware with no server and no network.

5.6 Does the epistemic signal help? A fair ablation

The preceding experiments show the epistemic modifier *fires*; this one tests whether it *helps*, isolating its contribution from the mere presence of an abstention option. We hold the system, the store, and the abstention mechanism fixed and toggle *only* the modifier: a **baseline** ranks with `use_epistemic=False` and an **epistemic** policy with `use_epistemic=True`; both then abstain when the top composite score falls below a threshold θ , which we sweep for each so neither is advantaged by a hand-picked value. The store holds 15 deliberately diverse facts per class (verified intact: 0 supersession events, 30/30 memories current, so there is no duplicate-collapse artifact): *trusted* (current, verified—should answer), *low-trust* (current, but marked CONTRADICTED/ UNCERTAIN with low confidence—should abstain), and *absent* (nothing relevant stored—should abstain). Correctness is value-based, independent of which memory is retrieved.

The decisive subset is *low-trust*, where a still-relevant memory scores high on pure relevance: the baseline confidently surfaces it, whereas the epistemic modifier multiplies its score by the status penalty and confidence floor, pushing it below θ . Table 3 reports a matched operating point ($\theta=0.60$, where *both* policies retain 100% answerable coverage); Figure 3 shows the full sweep and Figure 4 the low-trust subset.

Takeaway. At an identical threshold and identical 100% answerable coverage, folding stored reliability into ranking cuts low-trust confident-wrong answers from 33.3% to 0% (overall confident-wrong 11.1% $\rightarrow$ 0%; abstention F1 0.91 $\rightarrow$ 1.00). The effect is robust across the coverage-preserving band, not a single- θ artifact: the epistemic policy reaches 0% low-trust confident-wrong by $\theta=0.55$ while the baseline is still at 80% there. Because both policies share the same store and abstention rule, the gain is attributable to the epistemic signal itself, not to having an abstention option. The store is small ($n=15$ per class) and the low-trust conditions are constructed, so this isolates the read-path mechanism rather than demonstrating field prevalence (§7).

Table 3: Fair ablation: same store, only the epistemic modifier toggled, abstention threshold swept, reported at a matched operating point ($\theta=0.60$; both policies at 100% answerable coverage; $n=15$ per class; on-device, LLM-free). Lower is better for confident-wrong rates.

Metric	Baseline (relevance-only)	Epistemic
Answerable accuracy ($\uparrow$)	100%	100%
Low-trust confident-wrong ($\downarrow$)	33.3%	0.0%
Absent confident-wrong ($\downarrow$)	0.0%	0.0%
Overall confident-wrong ($\downarrow$)	11.1%	0.0%
Abstention F1 ($\uparrow$)	0.91	1.00

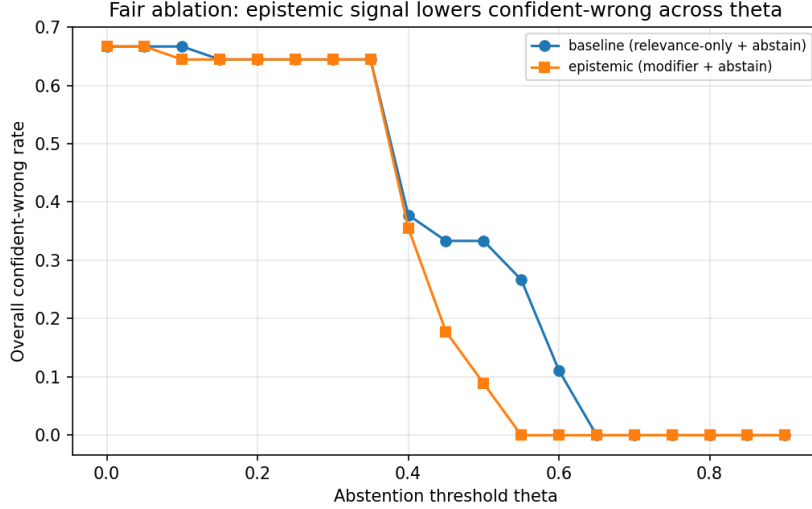


Figure 3: Confident-wrong rate versus the abstention threshold θ , for the relevance-only baseline and the epistemic policy on the identical store. The epistemic curve sits at or below the baseline throughout: the same retrieval, with stored reliability folded in, makes fewer confident errors.

6 Proposed Extensions

The epistemic state above also enables capabilities we specify but do not yet fully evaluate; we describe them precisely and release harnesses so they can be measured. (Grounded abstention—the most developed of these—is evaluated in §5.6.)

Belief propagation through consolidation. A memory derived from others (consolidation or synthesis) should inherit the *most conservative* status of its sources and a confidence that is a function of source agreement, with a DERIVED provenance edge. Provenance for derived memories is recorded today; principled status/confidence propagation through the consolidation engine is specified here and is partially realized.

External benchmarking. Our experiments characterize mechanics, not comparative accuracy. We release a benchmark runner that ingests each LongMemEval-S [15] chat history into Memory Layer, answers through the abstention policy, and emits predictions in the benchmark’s format for its official judge—reporting overall accuracy and, in particular, the *knowledge-update* and *abstention* slices, where an epistemic memory should help most. This is our recommended next step.

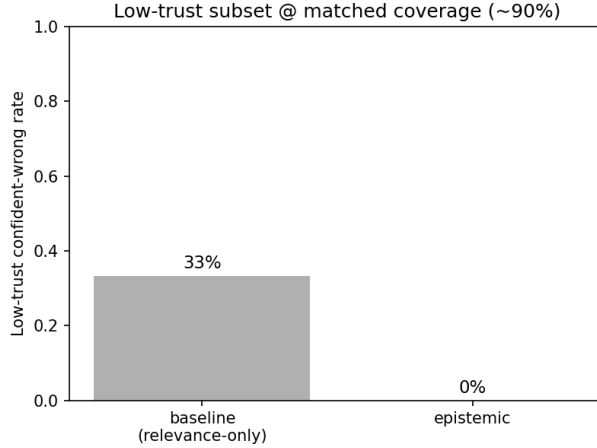


Figure 4: Low-trust subset at the matched operating point ($\theta=0.60$, 100% answerable coverage). The baseline confidently answers from memories the system itself marked contradicted/uncertain (33.3%); the epistemic policy demotes them below threshold and abstains (0%).

7 Limitations

- **No competitive QA benchmark yet.** We report no LoCoMo/LongMemEval accuracy or head-to-head comparison against Mem0/Zep/Letta; the runner exists but full, baseline-matched runs have not been done, so we make no accuracy-superiority claim.
- **Controlled, partly synthetic demonstrations.** §5.3–5.6 use small, constructed inputs to isolate mechanics, not field prevalence: in particular the fair ablation uses $n=15$ facts per class with constructed low-trust statuses, and its effect lives in a fairly narrow (though internally robust) θ band.
- **Confidence is heuristic, not calibrated.** The priors of Eq. (1) are hand-set; we do not show they match empirical correctness (a reliability-diagram/ECE study is future work).
- **Consolidation synthesis needs an LLM.** The fully LLM-free path is extractive; abstractive synthesis benefits from a (local) LLM.
- **Hand-set thresholds.** The duplicate cutoff (0.95), staleness window (14 days), and modifier constants (W_f , the penalties) are not tuned or learned.
- **Single machine.** Multi-user/distributed behavior is out of scope.

8 Conclusion

We presented a memory layer that pushes the *local*, *LLM-free* constraint as far as it will go and makes a memory’s self-knowledge operational: reliability governs retrieval, and the store audits itself, all on-device and characterized on consumer hardware. The contribution is integration and operationalization, not a new algorithm, and we characterize rather than claim to beat a benchmark. The unifying view is that contradiction handling, knowledge updates, provenance, self-audit, and abstention are one thing seen from different angles: operations on *belief*. We release all code and reproduction scripts, and identify external long-horizon evaluation as the clear next step.

Reproducibility and Availability

The system, the maintenance loop, the epistemic modifier, the self-audit, and a test suite of 147 unit and integration tests are released as open source, together with scripts that regenerate every figure and number in this paper (the characterization suite for §5.1–5.5, the fair-ablation

script for §5.6, and a runner for the LongMemEval benchmark). The on-device experiments are deterministic and require no API key. The repository URL will accompany the public release.

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